EX NO:1 BASIC NETWORKING COMMANDS OF WINDOWS AND LINUX

## Networking commands in windows and Linux

**Aim:**

To write a basic networking commands of windows and linux.

**command 1:**

window : ip config
Linux : ip wolder

Purpose : show your IP address and networking details.

Eg output :

wireless LAN adaption wifi:
IPV6 Address . . . . : 2406:7400:cc:4faf:2ffc
IPV4 Address . . . . : 192.168.68.9
subnet mask . . . . : 255.255.255.0

**command 2:**

window: ip config /all
Linux : ip address show

Purpose : shows complete network configuration.

sample output:

windows Ip configuration
Host Name . . . . : DESKTOP-KNBRGCV
Primary Dns suffix . . . . :
Node Type . . . . : Hybrid
IP Routing Enabled : No

**command 3:**

window : Ping google.com
Linux : Ping google.com
Purpose : checks if another device on website is reachable.

sample output:

Pinging google.com [142.250.77.110] with 32 bytes of dat
Reply from 142.250.77.110 bytes = 32 times = 5ms TTL = 11
Replay from 142.250.77.110: bytes = 32 times = 6ms TTL
Approximate round trip times in milli-sec:
Minimum = 5ms, max = 8ms, Average = 6ms.

**command 4:**

window: Tracert google.com
Linux : Tracert google.com
Purpose : shows the path packets take to a destination

sample output:

Tracing route to google.com [142.250.77.110]
over a maximum of 30 hops:

| | | | | |
|---|---|---|---|---|
| 1 | 1ms | 1ms | 2ms | 172.16.8.1 |
| 2 | 4ms | 3ms | 12ms | 137.97.163.129 |
| 3. | * | * | * | Request time out |

Trace complete.

command 5:

window : nslookup google.com
linux : nslookup google.com
Purpose : Finds the IP address of a domain
sample output :
server : unknown
Address : 172.16.8.1
Non-authoritative answer :
Name : Google.com

command 6:

window : netstat -an
Linux : netstat -tuln
Purpose : Display active network connections.
sample output :
Active connections

| Proto | Local address | Foreign address | state |
|---|---|---|---|
| TCP | 0.0.0.0.135 | 0.0.0.0 | LISTENING |
| TCP | 0.0.0.0.445 | 0.0.0.0 | LISTENING |
| TCP | 172.16.11.150.50156 | 4.123.25.40 | ESTABLISH |
| TCP | 172.16.11.150.61650 | 184.28.109.178 | TIME-WAIT |

COMMAND 7:

window : hostname
Linux : hostname
Purpose : display the computer name

sample output :
window → DESKTOP-KNBRGCD
Linux → Prathiksha-pc

command 9:

window : getmac
Purpose : show the mac address
sample output :

| Physical address | Transport Name |
|---|---|
| E8-80-88-8F-A5-D7 | Media Disconnected |
| DC-46-28-1F-52-ED | \Device\TCPIP_{4C28955EF-AAEC-4AAC-FCEB0F268B2} |

command 10:

Linux : route -n or iproute
purpose : shows routing table
sample output
default via 192.168.1.1 dev eth0
192.168.1.0/24 dev eth0 proto kernel scope
link src 192.168.1.100

Result :
Basic Networking command implemented successfully.